

Controls on the hydrogen isotopic composition of biogenic methane from high-latitude terrestrial wetlands

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[1] To investigate the controls on the δD isotopic composition of biogenic CH_4 in terrestrial wetlands, we collected a series of samples for $\delta D-H_2O$, $\delta D-CH_4$, $\delta^{13}C-CH_4$, and $\delta^{13}C-DIC$ (dissolved ΣCO_2) along a N-S transect across Alaska from 60°N to 70°N latitude from 7 to 15 August 2001. The $\delta D-H_2O$ and $\delta D-CH_4$ values varied from -108‰ to -161‰ and from -308‰ to -394‰ , respectively, from south to north and were significantly correlated, indicating the significant influence of the $\delta D-H_2O$ on the δD of terrestrial CH_4 collected along a latitudinal spatial gradient. Additionally, the apparent fractionation factors (α , $\alpha_{A-B} = R_A/R_B$, where $R = {}^{13}C/{}^{12}C$ or D/H) for $H_2O \rightarrow CH_4$ (α_D) and $DIC \rightarrow CH_4$ (α_C) varied from 1.26 to 1.42 and from 1.035 to 1.084, respectively, and were inversely correlated. We conclude that while $\delta D-H_2O$ is a critical factor controlling $\delta D-CH_4$ across latitudes, the CH_4 production mechanism is also responsible for variation in the $\delta D-CH_4$. If the isotopic values of the precursors of methane, H_2O , DIC , and organic matter are relatively constant, CH_4 produced by acetate fermentation will be enriched in $\delta^{13}C$ and depleted in δD relative to CH_4 produced via the CO_2 reduction pathway. Production mechanism variation will be particularly important in controlling variations in CH_4 isotopic composition with depth. The strong dependence of $\delta D-CH_4$ on the δD of environmental H_2O indicates that specific fields on plots of $\delta D-CH_4$ versus $\delta^{13}C-CH_4$ may not always accurately represent the isotopic composition of CH_4 produced by CO_2 reduction in northern wetlands. Because of this variation at high latitude, we assert that $\delta D-CH_4$ values cannot be associated with production mechanism in an absolute sense. The production mechanism effect is in addition to the effect of the $\delta D-H_2O$.

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1. Introduction

[2] Methane is the final end product of the anaerobic decomposition of organic matter. It is formed by two main pathways, acetate fermentation and reduction of CO_2 with hydrogen. Variation in these two main mechanisms of CH_4 production has been hypothesized to be associated with inverse or antipathetic shifts in δD and $\delta^{13}C$ of CH_4 [Whiticar *et al.*, 1986; Schoell, 1988; Burke *et al.*, 1988a, 1988b; Whiticar, 1999; Hornibrook *et al.*, 1997; Chanton *et al.*, 2005]. It is often stated that fermentation of acetate will result in CH_4 that is ^{13}C enriched and D depleted relative to CH_4 produced via the CO_2 reduction pathway. Whiticar *et al.* [1986] defined somewhat distinct isotope compositional fields on a plot of $\delta D-CH_4$ versus $\delta^{13}C-CH_4$ resulting from the different biogenic pathways and from thermogenic and geothermal origin. However, a portion of the δD variation is

due to variation in the precursor for CH_4 hydrogen, which is in part H_2O . As discussed by Waldron *et al.* [1999], part of the reason for the δD distinction between acetate-formed CH_4 versus CO_2 reduction CH_4 is because samples that were representative of the CO_2 reduction pathway were often from the marine environment where the $\delta D-H_2O$ is close to 0‰ . Methane representative of acetate fermentation was more likely sampled from terrestrial environments where the $\delta D-H_2O$ was somewhat depleted relative to marine H_2O [Waldron *et al.*, 1999]. For terrestrial environments at high latitudes, the effect will be more acute [Waldron *et al.*, 1999].

[3] While the usefulness of $\delta^{13}C-CH_4$ as an indicator of methanogenic pathway has been recently demonstrated [e.g., Conrad *et al.*, 2002; Sugimoto and Wada, 1993], the variation of $\delta D-CH_4$ with shifts in the acetate fermentation and CO_2 reduction pathways has been called into question [Waldron *et al.*, 1999, 1998b; Sugimoto and Wada, 1995]. In several elegant experiments, investigators who have incubated organic matter of differing reactivity with H_2O of differing δD isotopic composition [Sugimoto and Wada, 1995; Waldron *et al.*, 1998a] have failed to observe shifts in the δD of CH_4 with changing production mechanism. What

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does appear to control the δD of CH_4 in these incubation studies is the δD of H_2O . The $\delta D-CH_4$ values fall very close to a single line ($r = 0.986$) when plotted against the $\delta D-H_2O$ used in the experiment [Waldron *et al.*, 1999].

[4] Waldron *et al.* [1999] also plotted 51 cogenetic pairs of $\delta D-H_2O$ and $\delta D-CH_4$ from environmental samples from a variety of environments across latitude. They obtained a significant relationship between $\delta D-H_2O$ and $\delta D-CH_4$ and concluded that the isotopic composition of formation H_2O was the primary control on the $\delta D-CH_4$. They observed, however, that there was more variation for $\delta D-CH_4$ than for $\delta D-H_2O$, which might be caused by other processes. One of the purposes of this paper is to determine if variations in CH_4 production pathways might be one of these processes. Furthermore, the data of Waldron *et al.* [1999] was weighted toward more tropical and temperate areas. The authors suggested a need for additional data at higher latitudes.

[5] Chanton *et al.* [2005] presented results from field studies ranging from tropical to boreal regions and provided strong evidence that CH_4 produced from acetate fermentation is relatively ^{13}C enriched and δD depleted relative to CH_4 produced from reduction of CO_2 . However, they pointed out that the relationship only holds if precursor isotopic values ($\delta^{13}C-DIC$ and $\delta D-H_2O$) are relatively constant across the spatial or temporal gradients across which the measurements are made.

[6] In this paper we will examine field data to provide evidence that although precursor H_2O isotopic composition is a primary factor controlling the $\delta D-CH_4$, variations in CH_4 production mechanisms are a second controlling factor. Therefore along a N-S transect across Alaska we measured δD and $\delta^{18}O$ of H_2O , $\delta^{13}C$ and δD of CH_4 and $\delta^{13}C$ of DIC. Our first objective was to determine if fractionation of $\delta D-CH_4$ varies with CH_4 production mechanism by examining the variation of apparent fractionation factors for the pairs $\delta^{13}C-DIC$ and $\delta^{13}CH_4$ and $\delta D-H_2O$ and $\delta D-CH_4$. To accomplish this we sampled semisynoptically across a horizontal gradient at different scales and synoptically along a depth gradient. Methane production mechanisms have been observed to vary as a function of depth in northern wetlands [Hornibrook *et al.*, 1997; Chasar *et al.*, 2000a, 2000b; Chanton *et al.*, 2005]. Our second objective was to extend the observations of Waldron *et al.* [1999] by collecting additional data from a transect between 60 and 70 degrees north in Anchorage, Fairbanks and along the pipe line road in Alaska up to Deadhorse.

2. Methods

[7] Samples were collected from bogs and fens in Alaska along a latitudinal gradient, from boreal to polar regions (60.0 N to 69.5 N) and across a depth gradient at a single site. Sampling occurred from 7 to 15 August 2001 and represents a synoptic data set, not a seasonal study. Replicate samples of pore water were collected from depths ranging from 10 to 50 cm with a “swamp sucker,” which is a quarter inch stainless steel tube with holes drilled at the base combined with a 50 mL syringe. At Turnagain Bog, near Anchorage Alaska (60°N latitude) in addition to shallow swamp sucker samples, additional samples were collected at depths ranging from 1 to 2 meters using a piezometer and a peristaltic pump. DIC (Dissolved Inor-

ganic Carbon or $\sum CO_2$), samples were filtered through glass fiber filters and 10 mL injected into 25 mL evacuated serum vials fitted with butyl rubber septa. DIC samples were stored frozen and upside down prior to analysis and then acidified with 0.3 mL of degassed 30% H_3PO_4 and brought to ambient pressure through an open split tube with nitrogen for testing. Small subsamples (20 μL) of the headspace from the DIC samples were run via direct injection on the GC-IRMS (HP-Finnigan Delta S) to determine $\delta^{13}C-DIC$. For $\delta^{13}C-CH_4$ and $\delta D-CH_4$ analysis, pore water was injected into evacuated 500 mL Qorpak brown glass bottles, containing 1.5 g KOH and stored upside down. Storage tests indicate that samples keep for over a year in this manner. Samples of H_2O for isotopic analysis were placed within hypovials and stoppered with a needle fitted through the septae to allow a sample to be collected with no headspace. The needle was removed after the stopper was inserted.

[8] The headspaces of the CH_4 samples were pressurized to one atmosphere with an open split tube and equilibrated with the dissolved phase. Then a small portion of each headspace was analyzed via direct injection into the GC-IRMS (Finnigan Delta S) to determine $\delta^{13}C-CH_4$ and concentration. Methane from the rest of the headspace was removed with a helium stream and combusted over copper oxide at 800°C and the resultant H_2O and CO_2 cryogenically trapped. The H_2O from the CH_4 combustion was isolated using an ethanol slush trap and transferred to glass tubes containing zinc shavings from Indiana University, which were then flame sealed. The tubes were heated to 500°C for 30 minutes to react the H_2O with the zinc and the resultant H_2 was analyzed for $\delta D-CH_4$ using the dual inlet system of the Finnigan MAT IRMS [Coleman *et al.*, 1982]. $\delta D-CH_4$ was determined relative to Standard Mean Ocean Water (SMOW) using gases that were calibrated against standard waters (V-SMOW, SLAP). Precision for δD measurements on the IRMS is 0.3‰ while overall precision of replicate field samples is 6 ‰ [Lombardi, 1999]. Samples of surface and pore water were sent to Waterloo University for analysis to determine the isotopic signatures of the pore water, $\delta D-H_2O$ and $\delta^{18}O-H_2O$. The $\delta^{13}C$ values were determined relative to VPDB in ‰. Lab standards were calibrated against NIST standards. Precision for $\delta^{13}C$ measurements was $\pm 0.2\text{‰}$ on the basis of repeated measurements of a laboratory working standard.

3. Data Analysis and Calculations

[9] Replicate samples were analyzed to discern the mean $\delta^{13}C-CH_4$, $\delta^{13}C-DIC$, $\delta D-CH_4$, and $\delta D-H_2O$ for each sample location and then an overall mean for each parameter for each site. From these data, the apparent fractionation factors [Chasar *et al.*, 2000a; Hornibrook *et al.*, 1997; Whiticar *et al.*, 1986] for $^{13}C/^{12}C$ and D/H (α_C and α_H), can be determined for each location and site. The apparent fractionation factors for $CO_2 \rightarrow CH_4$ (α_C) and $H_2O \rightarrow CH_4$ (α_D) [Chasar *et al.*, 2000a; Hornibrook *et al.*, 1997; Whiticar *et al.*, 1986] are calculated as

$$\alpha_C = \frac{\delta^{13}C - DIC + 1000}{\delta^{13}C - CH_4 + 1000} \quad (1)$$

Table 1. Isotopic Composition of Water Samples Versus Latitude^a

Sample	Latitude	$\delta^{18}\text{O}\text{‰}$	$\pm$	Mean $\delta\text{D}\text{‰}$	$\pm$
Turnagain Bog Site 1, 20cm	60.16	-14.50		-115.2	0.2
Turnagain Bog Site 2, 35cm	60.16	-13.55		-112.0	0.1
Turnagain Bog Site 3, 35cm	60.16	-16.30		-124.7	0.4
Turnagain Bog Site 4, 20cm	60.16	-16.23		-122.6	0.3
Turnagain Bog Site 5, 35cm	60.16	-15.47		-119.8	0.2
Turnagain Bog Site A, 35 cm	60.16	-14.19	0.02	-112.3	0.2
Turnagain Bog Site B, surface water	60.16	-13.50		-108.3	0.5
Turnagain Bog Site B, 50 cm	60.16	-15.76		-124.5	0.2
Turnagain Bog Site B, 100 cm	60.16	-16.03		-124.9	0.1
Turnagain Bog Site B, 150 cm	60.16	-16.23		-126.4	0.2
Turnagain Bog Site B, 200 cm	60.16	-16.27		-123.1	0.0
Fen 61, surface water	61.00	-15.13	0.10	-123.5	0.0
Fen 62, surface water	62.81	-14.30	0.05	-115.0	0.4
Fen 62, Sphagnum, 20 cm	62.81	-17.65		-138.4	0.3
Fen 62, Carex, 35 cm	62.81	-15.37		-125.8	0.1
Fen 63, surface water	63.00	-15.92		-129.0	0.5
Bonanza Creek Mix, surface water	64.70	-18.06		-139.9	0.2
Bonanza Creek Mix, 27 cm	64.70	-18.71	0.06	-145.0	0.6
Beaver Sports Hummock, 10cm	64.86	-18.38	0.20	-141.9	0.3
Beaver Sports Hollow, 10 cm	64.86	-18.41		-144.9	0.2
Smith Lake surface water	64.87	-13.41		-128.8	0.7
Smith Lake Carex, 40cm	64.87	-14.59		-136.4	0.0
UAF Bog, surface water	64.87	-18.53		-143.7	0.2
UAF bog, 25cm	64.87	-18.65	0.03	-145.0	0.0
Discontuous Permafrost, Surface water	66.09	-15.46		-131.6	0.2
Discontuous Permafrost Sphagnum, 30cm	66.09	-18.77		-154.2	0.4
Discontuous Permafrost Carex, 30cm	66.09	-19.79		-158.8	0.4
Fen 67, surface	67.48	-20.51		-160.7	0.4
Fen 67, 35cm	67.48	-20.51		-160.7	0.4
Galbraith Lake Slough, 35cm	68.48	-17.77		-145.1	0.7
Galbraith Lake Hummock, 35cm	68.48	-17.83		-146.6	0.1
Imnaviat Creek Carex, 35 cm	68.62	-20.25		-154.4	0.3
Imnaviat Creek Sphagnum, 30cm	68.62	-19.76		-149.8	0.2
Pipeline 70 Fen Carex, 25 cm	69.33	-19.11		-147.6	0.0
Franklin Bluff Fen, 45cm	69.80	-16.48		-134.8	0.2

^aSites at Turnagain Bog were roughly 10 meters apart. The $\pm$ represents 1/2 of the range of analytical replication of a single sample.

$$\alpha_D = \frac{\delta\text{D} - \text{H}_2\text{O} + 1000}{\delta\text{D} - \text{CH}_4 + 1000} \quad (2)$$

[10] Changes in the apparent α s have been hypothesized to represent changes in relative importance of the production mechanism (see discussion by *Whiticar* [1999]); larger α_C and smaller α_D are typical of CO_2 reduction while smaller α_C and larger α_D values are typical of acetate fermentation [*Whiticar et al.*, 1986; *Whiticar*, 1999; *Sugimoto and Wada*, 1993; *Conrad et al.*, 2002]. We refer to these factors as apparent, because while CO_2 and H_2O are the precursors for CH_4 formed from CO_2 reduction, they are not the immediate precursors for CH_4 formed from acetate fermentation. Nonetheless, variations in apparent fractionation factors are interpreted to represent variations in CH_4 production mechanism [*Conrad et al.*, 2002; *Chasar et al.*, 2000a; *Hornibrook et al.*, 1997, 2000a, 2000b].

4. Results

[11] The isotopic signatures of the water ($\delta\text{D}-\text{H}_2\text{O}$ and $\delta^{18}\text{O}-\text{H}_2\text{O}$) are reported in Table 1. Replicate analytical results are reported for every δD measurement and at intervals for $\delta^{18}\text{O}$. The surface and pore water samples fall along a line (Figure 1) with a slope of $6.9 (\pm 0.5)$ and an intercept of $-16.6 (\pm 7.7)$. The global meteoric water line has a slope of 8 with an intercept of +9 [*Hoefs*, 1997]. The

$\delta\text{D}-\text{H}_2\text{O}$ decreases with increasing latitude, but shows a slight increase approaching 70°N and the Arctic Ocean (Figure 2). Pore water samples were depleted in δD and $\delta^{18}\text{O}$ relative to surface water samples by an average of $11.2 \pm 9.8\text{‰}$ and $1.6 \pm 1.5\text{‰}$ (error represents 1 standard deviation), presumably due to surface enrichment by evaporation.

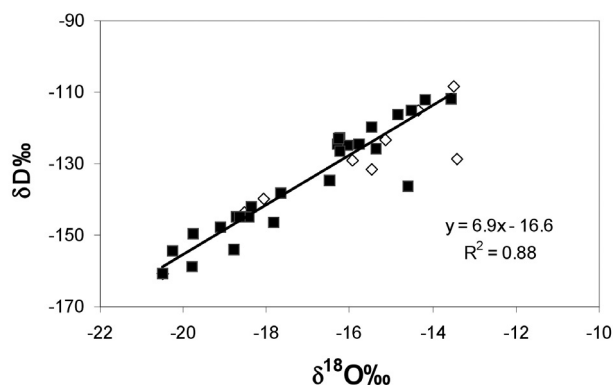


Figure 1. A plot of $\delta\text{D}-\text{H}_2\text{O}$ versus $\delta^{18}\text{O}-\text{H}_2\text{O}$ indicates that both isotopes are affected by the same processes, i.e., evaporation, transpiration, and precipitation. The correlation is significant, $p < 0.005$. Filled squares represent pore water samples while open diamonds are surface water samples.

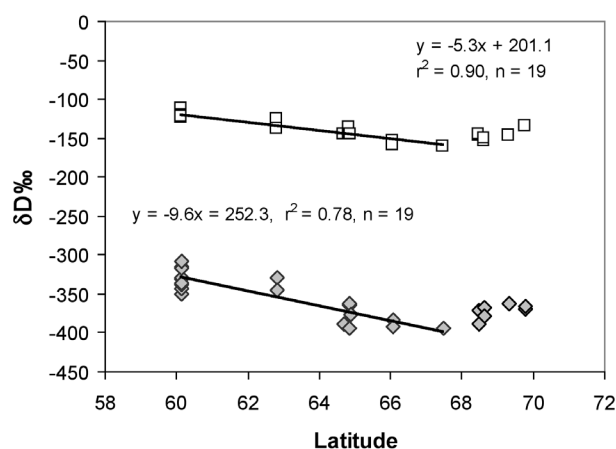


Figure 2. A plot of the average isotopic signatures, $\delta\text{D-CH}_4$ (diamonds) and $\delta\text{D-H}_2\text{O}$ (squares), of the pore water for each Alaska site/depth versus the latitude of the site. Note the increase in δD of both water and methane approaching 70 degrees north latitude and the coast of the Arctic Ocean. Correlation of both factors with latitude from 60 to 68 degrees is significant, $p < 0.005$.

[12] The $\delta^{13}\text{C}$ and δD mean values for duplicate sample analysis of the dissolved gases CH_4 and DIC are reported in Table 2. The $\delta^{13}\text{C}$ of DIC ranged from -19 to $+3\text{‰}$ driven by the importance of CH_4 production and possibly acetogenesis. The most enriched DIC- $\delta^{13}\text{C}$ values were found at depth in Turnagain Bog and at the discontinuous permafrost sites. Other than depth trends, there was no consistent pattern in DIC data, and there was no trend with latitude. Methane $\delta^{13}\text{C}$ ranged from -50‰ to -86‰ . Interestingly

both extremes of the range were obtained from *Carex* sites (Table 2). The $\delta\text{D-CH}_4$ values for these two sites were $-362 \pm 5\text{‰}$ and $-346 \pm 12\text{‰}$, respectively, so CH_4 oxidation is not immediately obvious as an explanation for the $\delta^{13}\text{C}$ variation observed. Had CH_4 oxidation driven the differences in $\delta^{13}\text{C}$ between these two sites, the δD values would likely have covaried [Coleman *et al.*, 1981]. The mean $\delta\text{D-CH}_4$ varied from -394‰ to -308‰ . As observed for the $\delta\text{D-H}_2\text{O}$, $\delta\text{D-CH}_4$ decreased with latitude and similarly exhibited a small increase approaching 70°N and the Arctic Ocean (Figure 2).

[13] A plot (Figure 3a) of $\delta\text{D-CH}_4$ versus $\delta\text{D-H}_2\text{O}$ for the Alaska data yields a significant correlation at the 99% confidence level indicating that δD of H_2O controls a large portion of the variation in $\delta\text{D-CH}_4$. The Alaska transect data extend the Waldron *et al.* [1999] data set as shown in Figure 3b. The overall fit to the entire data set has a slope of 0.54 and an intercept of -290 . The linear regression fit to the data of Waldron *et al.* has a slope of 0.74 and an intercept of -284.3 . The Alaska data have a steeper slope (1.55) and a more positive intercept (-145.4) relative to the other relationships. Waldron *et al.* [1999] obtained a slope of 0.68 and an intercept of -284 but used a weighted linear regression using the standard error of the mean $\delta\text{D-CH}_4$ for each site, whereas we used the actual values.

[14] One objective of this paper is to extend the observations of Waldron *et al.* [1999] to include more northern latitude data in the examination of the relationship between $\delta\text{D-CH}_4$ and $\delta\text{D-H}_2\text{O}$ across the globe. Clearly, a large portion of the variation of the $\delta\text{D-CH}_4$ data is explained by the variation of the $\delta\text{D-H}_2\text{O}$ results (Figure 3a). To the extent that the proportion of the variability in the y values is explained by a straight line fit to the x values, the data indicate that roughly 70% of the variability of $\delta\text{D-CH}_4$ can

Table 2. $\delta^{13}\text{C-CH}_4$, $\delta\text{D-CH}_4$, $\delta^{13}\text{C-DIC}$ and $\delta\text{D-H}_2\text{O}$ Data for Each Site^a

Site	Latitude	depth, cm	$\delta^{13}\text{C CH}_4$	$\pm$	$\delta^{13}\text{C DIC}$	$\pm$	$\delta\text{D CH}_4$	$\pm$	$\delta\text{D H}_2\text{O}$
Turnagain Bog Site 1	60.16	20	-64.0	0.4	-7.1	0.3	-350.7	2	-115.2
Turnagain Bog Site 3	60.16	35	-71.9	0.6	-23.1	0.2	-331.4	2	-124.7
Turnagain Bog Site 4	60.16	20	-75.0	0.1	-8.9	0.8	-328.9	1	-122.6
Turnagain Bog Site 5	60.16	35	-63.9	0.5	-7.2	0.1	-343.0	10	-119.8
Turnagain Bog Site A	60.16	35	-73.0	1.0	-2.9	0.1	-338.8	4	-112.3
Turnagain Bog Site B	60.16	50	-70.1	0.5	-4.3	0.4	-335.6	5	-124.5
Turnagain Bog Site B	60.16	100	-69.3	0.4	-2.2	0.9	-314.7	7	-124.9
Turnagain Bog Site B	60.16	150	-69.3	0.7	1.1	0.3	-317.7	1	-126.4
Turnagain Bog Site B	60.16	200	-70.5	0.7	2.9	0.5	-307.9	10	-123.1
Fen 62 Carex	62.81	35	-82.2	0.2	-7.6	1.2	-329.9	3	-125.8
Fen 62 Sphagnum	62.81	20	-86.0	0.2	-13.3	0.1	-345.8	12	-138.4
Bonanza Mix	64.70	27	-55.1	0.1	-16.4	0.7	-389.5	2	-145.0
Beaver Sports Hollow	64.86	10	-56.6	0.1	-18.0	0.1	-394.1	7	-144.9
Beaver sports Hollow	64.86	40	-56.9	1.0	-14.0	0.3	-365.0	15	*
Smith Lake carex	64.87	40	-50.5	0.3	-19.1	0.1	-362.4	5	-136.4
UAF Bog	64.87	25	-52.3	0.4	-15.1	0.2	-377.1	10	-145.0
Discont. Perm Sphagnum	66.09	30	-62.8	0.1	-0.2	0.5	-384.1	5	-154.2
Discont. Perm Carex	66.09	30	-65.5	0.3	2.1	0.4	-392.7	3	-158.8
Fen 67	67.48	35	-66.2	1.3	-8.2	0.1	-394.0	3	-160.7
Galbraith Lake Hummock	68.48	35	-67.2	0.2	-7.3	0.3	-371.7	5	-146.6
Galbraith Lake Slough	68.48	35	-62.9	0.2	-5.9	0.1	-388.2		-145.1
Imnaviat Creek Carex	68.62	35	-62.3	0.3	-15.2	1.3	-367.0	13	-154.4
Imnaviat Creek Sphagnum	68.62	30	-65.9	0.9	-19.1	2.7	-379.0	2	-149.8
Pipeline 70 Fen	69.33	25	-57.8	0.1	-12.4	0.1	-363.3	14	-147.6
Franklin Bluff fen	69.80	25	-58.1	0.1	-9.0	0.5	-369.0	4	*
Franklin Bluff fen	69.80	45	-62.8	0.8	-5.5	0.0	-366.0	5	-134.8

^aAsterisk indicates that $\delta\text{D-H}_2\text{O}$ data was used from a pore water sample from the same site but a different depth for regression analysis and alpha calculations. The $\pm$ represents 1/2 of the range for replication of duplicate samples.

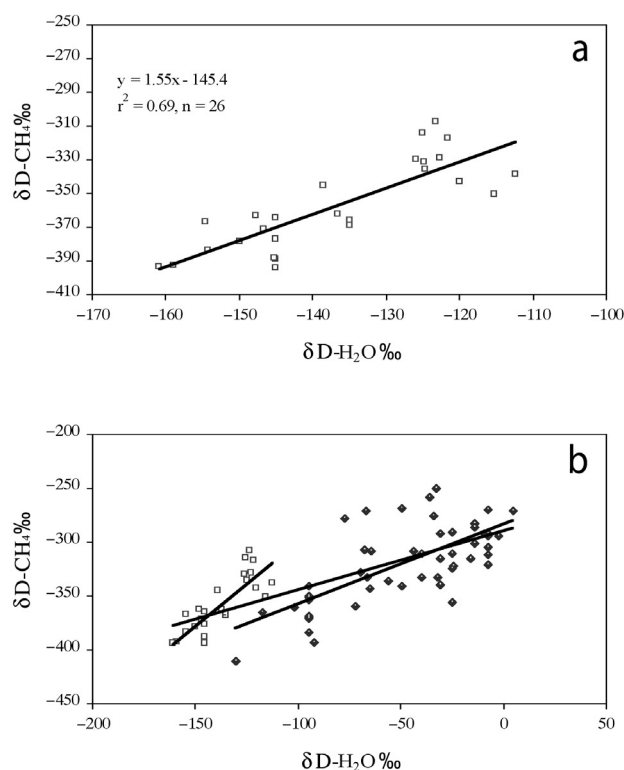


Figure 3. (a) Trend of $\delta D-CH_4$ ‰ versus $\delta D-H_2O$ ‰ in samples from the Alaska transect, standard error of slope, 0.46, intercept 29.91, $p < 0.005$. (b) Trend of $\delta D-CH_4$ ‰ versus $\delta D-H_2O$ ‰, Alaska data, as in Figure 3a (squares), with data from Waldron et al. [1999] (diamonds) included. Our line fit to the Waldron data is $y = 0.74 (\pm 0.11)x - 284.3 (\pm 6.3)$; $r^2 = 0.50, n = 51, p < 0.005$. The line fit to the entire data set is $y = 0.54 (\pm 0.05)x - 290.1 (\pm 5.3)$, $r^2 = 0.53, n = 77, p < 0.005$.

be explained by $\delta D-H_2O$. We will also examine the data to discern if a portion of the variation of $\delta D-CH_4$ about the line fit to the $\delta D-H_2O$ is due to variation in CH_4 production mechanism.

5. Discussion

[15] Generally, the slope of the line of $\delta D-CH_4$ fit to $\delta D-H_2O$ has been interpreted as being indicative of the CH_4 production mechanism, with CH_4 generated purely from CO_2 reduction having a slope of 1 because all of the CH_4 hydrogen is thought to be derived from coexisting H_2O . Acetate fermentation is thought to have a slope of 0.25 as only 1 hydrogen has been considered to be derived from environmental H_2O while the other 3 have been thought to be transferred intact from the acetate methyl group [Whiticar, 1999; Whiticar et al., 1986]. Strict interpretation of the overall results (Figure 3b) in light of this paradigm would suggest a 60–40 mix of acetate fermentation and CO_2 reduction in the generation of global terrestrial CH_4 as included in the data set. However, the picture is probably not so simple as it initially appeared to be [Waldron et al., 1999]. Recent evidence has suggested

that there may be an exchange of hydrogen between the acetate methyl group and H_2O in the final stages of methanogenic acetate metabolism [de Graaf et al., 1996]. Presumably this isotopic exchange is accompanied by an isotopic fractionation but the magnitude of this effect has not been quantified, as pointed out by Hornibrook et al. [1998].

[16] Hydrogen concentration has also been suggested as a factor that may influence the isotopic composition of H in CH_4 [Burke, 1993; Sugimoto and Wada, 1995], and this factor may be particularly responsible for the lack of agreement between incubation studies and surveys of natural wetlands [Chanton et al., 2005]. Burke [1993] proposed that hydrogen concentration could affect the fractionation of H between H_2O and CH_4 , resulting in greater fractionation at higher H_2 concentrations but not affecting the $\delta^{13}C$ of CH_4 . Recently, Valentine et al. [2004] observed that microbial growth rate can have a profound effect on fractionation (α_D) between H_2O and CH_4 during CO_2 reduction, thus influencing the δD of CH_4 . Furthermore environmental H_2O δD variation affects the δD of organic matter and thus should affect the δD of the acetate methyl group. Given these uncertainties, it is best not to interpret the slope value too strictly. It can be suggested on the basis of Figures 3a and 3b that 50 to 70% of the variation in $\delta D-CH_4$ is directly related to variation in δD of environmental H_2O .

[17] It is interesting that the Alaska data set indicated a slope of 1.5 (Figure 3a), greater than what would have been expected for CO_2 reduction alone. This finding could be due to variation with production mechanism with latitude in our data set. Lower-latitude ($60^\circ N$) data were dominated by samples from Turnagain Bog where acetate is known to accumulate in interstitial water and not form CH_4 [Hines et al., 2001; Duddelston et al., 2002]. In general studies have been interpreted to indicate that CH_4 production via the acetate pathway is less consequential in high-latitude *Sphagnum*-dominated bogs [Lansdown et al., 1992; Chasar et al., 2000a, 2000b; Chanton et al., 2005]. Variation in radiocarbon distributions between DOC, CH_4 , and DIC further support the idea that respiration modes fundamentally differ in bogs and fens [Chasar et al., 2000b]. Turnagain bog samples varied from -115 ‰ to -123 ‰ in $\delta D-H_2O$ isotopic composition, and thus represented the right side of the graph in Figure 3a. Nonetheless the Turnagain data fit the line quite well. As most of the sites we sampled were represented by only 1 or 2 data, we investigated reducing the number of surface Turnagain data points to only two randomly chosen surface samples, samples 1 and 3, which are fairly representative of the other surface samples. This reduction changed the line fit to the data little, to $y = 1.27x - 188.45$, $r^2 = 0.59$. Thus the possible overrepresentation of Turnagain Bog as a lower-latitude (60°) sampling site is not responsible for the observed pattern.

[18] Addition of the Alaska data to the Waldron et al. [1999] data set has the effect of reducing the slope of the line of global water versus global CH_4 , converging the field data with the laboratory data. The new slope is more similar to the slope of the plot of $\delta D-CH_4$ versus $\delta D-H_2O$ obtained from the incubation results of Waldron et al. [1998a], Sugimoto and Wada [1995], and Schoell [1980] as plotted by Waldron et al. [1999], which had a slope of 0.44 and an intercept of -321 .

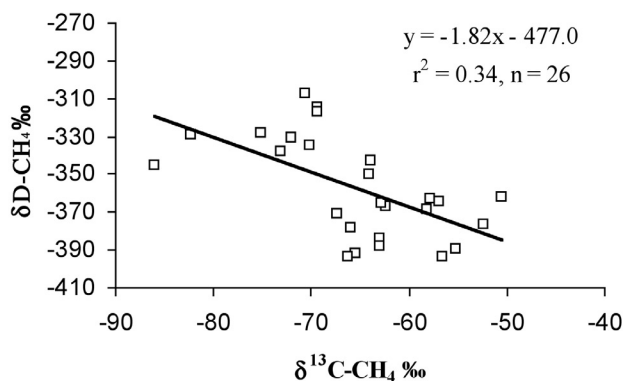


Figure 4. Plot of $\delta\text{D-CH}_4$ versus $\delta^{13}\text{C-CH}_4$ for Alaska transect samples. While there is a significant inverse correlation between the values, a portion of this variation is due to variation in the precursors, $\delta^{13}\text{C-DIC}$ and $\delta\text{D-H}_2\text{O}$. Standard error on slope is 0.52 and on intercept is 34.2; $p < 0.005$.

[19] The intercept of the line fit to the $\delta\text{D-CH}_4$ fit to $\delta\text{D-H}_2\text{O}$ has been interpreted as being indicative of the fractionation between water H and the resulting methane H. The line fit to marine data, which is thought to be indicative solely of CO_2 reduction, has an intercept of -160‰ , while the intercept of various lines fit to acetate fermentation vary from -300‰ to -370‰ [Whiticar, 1999]. Our Alaska data had a more positive intercept than the overall data set, which might be interpreted as due to a greater influence of CO_2 reduction at higher latitudes because of the inhibition of the acetate fermentation pathway in *Sphagnum*-dominated bogs. However, the steeper slope of the line clearly influenced the intercept so this interpretation is somewhat suspect.

[20] We next evaluate the hypothesis that departures from the linear fit observed in Figure 3a are related to variations in the CH_4 production mechanism. We will assume that variation in the signature of $\delta^{13}\text{C-CH}_4$ indicates variation in CH_4 production mechanism, as discussed by Conrad et al. [2002] [see also Sugimoto and Wada, 1993; Waldron et al., 1998a], since samples were collected from depths that were for the most part 25 cm below the surface and thus should be relatively free of CH_4 oxidation as was observed for northern wetlands by Popp et al. [1999]. When $\delta\text{D-CH}_4$ is plotted versus $\delta^{13}\text{C-CH}_4$, an inverse correlation results (Figure 4), consistent with this assumption. If $\delta^{13}\text{C-CH}_4$ was controlled by CH_4 oxidation, a positive correlation between $\delta^{13}\text{C}$ and δD of CH_4 would have been observed [Coleman et al., 1981; Chanton et al., 2005]. Additionally, the finding of a negative correlation in Figure 4 is consistent with the hypothesis that some of the variation in δD that we observed across our transect was due to variation in CH_4 production mechanism. In Figure 4, there appear to be two clusters of data, those on the left where CO_2 reduction is more important, and those on the right where acetate fermentation is more dominant. The left cluster data are from latitudes below 63°N and include data from Turnagain bog, where incubation studies have clearly demonstrated that CO_2 reduction is the dominant production mechanism.

The right-side data are from above 64°N . We suspect that this segregation of the data with latitude is serendipitous, and does not imply that there are variations in CH_4 production pathways with latitude on this scale. On larger latitudinal scales, however, for example in *Sphagnum*-dominated wetlands which are more often found in northern latitudes, one is likely to observe that CO_2 reduction is the dominate methane production mechanism. This finding has been observed in both isotopic and incubation studies [Landsdown et al., 1992; Chasar et al., 2000a, 2000b; Hines et al., 2001; Duddleston et al., 2002].

[21] However, some portion of the variation in $\delta\text{D-CH}_4$ in Figure 4 is caused by the variation in $\delta\text{D-H}_2\text{O}$ with latitude (Figure 3a). To remove the effect of precursor variation, we can determine if the apparent fractionation factors (α_C and α_D) for $\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$ vary inversely with respect to $\delta^{13}\text{C-DIC}$ and $\delta\text{D-H}_2\text{O}$. Plots of $\delta^{13}\text{C-CH}_4$ and $\delta^{13}\text{C-DIC}$ and $\delta\text{D-CH}_4$ and $\delta\text{D-H}_2\text{O}$ are shown in Figures 5a and 5b, respectively. A wide range of α_C values (equation (1)) was observed, from 1.035 to 1.084. The α_{DS} (equation (2)) ranged from 1.26 to 1.42. While the α_{CS} plot perpendicular to the lines of constant α , the α_{DS} are at more of an angle, possibly because of the variation of $\delta\text{D-H}_2\text{O}$ with latitude.

[22] The α_{DS} are plotted versus α_{CS} in Figure 6, and a significant inverse relationship is obtained ($p > 0.025$). This

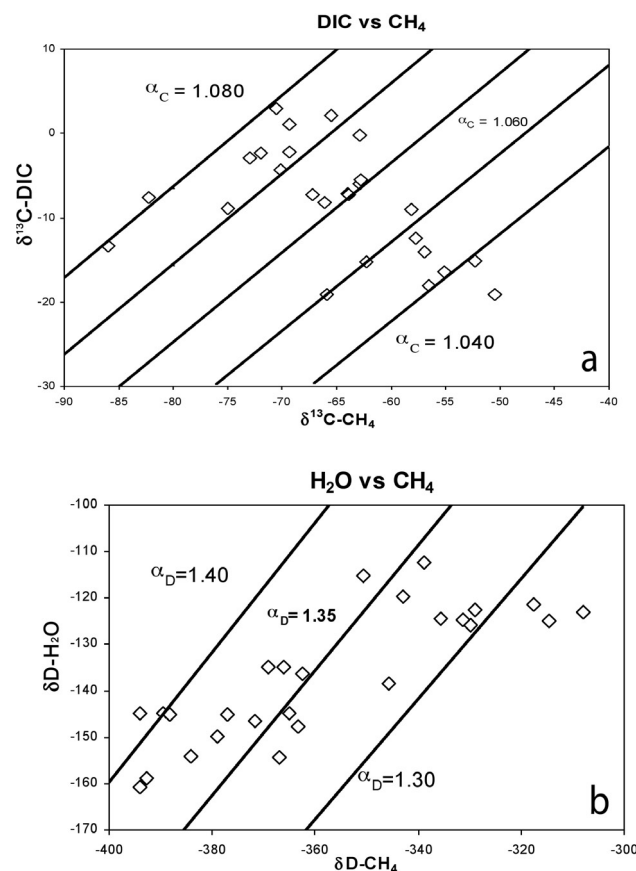


Figure 5. (a) Cross plot of $\delta^{13}\text{C-DIC}\text{‰}$ versus $\delta^{13}\text{C-CH}_4\text{‰}$ in pore waters. Lines depict varying values of α_C . (b) Cross plot of $\delta\text{D-H}_2\text{O}\text{‰}$ versus $\delta\text{D-CH}_4\text{‰}$ in pore waters. Lines depict varying values of α_D .

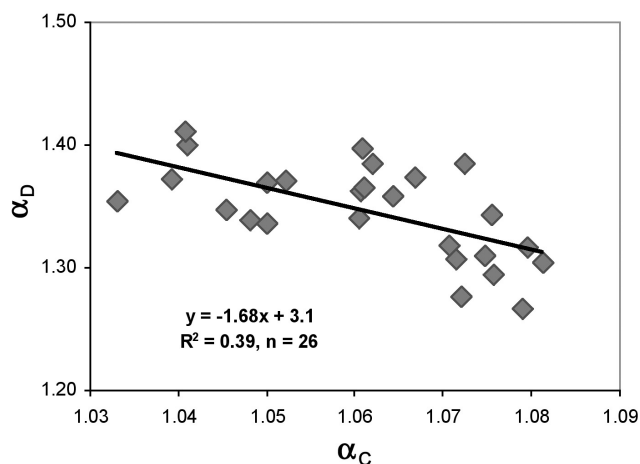


Figure 6. Plots of the α_D and α_C of methane produced in peatland sites along a transect in Alaska. Plot shows a significant antipathetic relationship ($p < 0.005$) between α_D and α_C presumably resulting from shifts in production mechanism. Standard error on slope, 0.43; intercept, 0.45.

means that as α_C increases α_D decreases and vice versa. Assuming that the variation in $\delta^{13}\text{C}-\text{CH}_4$ is due to variation in CH_4 production mechanism, then this finding is consistent with the hypothesis that $\delta\text{D}-\text{CH}_4$ varies with production mechanism as well, in an opposite manner to $\delta^{13}\text{C}-\text{CH}_4$. Thus it appears that variation in production mechanism is a second factor controlling the $\delta\text{D}-\text{CH}_4$. As with any correlation, we cannot address the causes of the variation, we can only note that our data are consistent with its occurrence. Again, it may be argued that the data overrepresent Turnagain Bog. Omitting all but two of the surface samples and refitting the linear regression did not change the slope or the correlation coefficient ($p > 0.025$).

[23] Our results indicate that $\delta\text{D}-\text{H}_2\text{O}$ is a primary factor controlling $\delta\text{D}-\text{CH}_4$ across latitudes, but that CH_4 production mechanism also affects $\delta\text{D}-\text{CH}_4$. If precursor isotopic values are relatively constant, CH_4 produced by acetate fermentation may be enriched in ^{13}C and depleted in D relative to CH_4 produced via the CO_2 reduction pathway. For depth profiles at one site, variation in production mechanism may be the dominant feature controlling δD and $\delta^{13}\text{C}$ variation in CH_4 [Hornibrook *et al.*, 1997].

[24] If CH_4 precursor isotope values vary, then variation in production mechanisms can be discerned by variation in apparent α values. Given the strong dependence of $\delta\text{D}-\text{CH}_4$ on the δD of environmental water, it seems to us that specific fields on plots of $\delta\text{D}-\text{CH}_4$ versus $\delta^{13}\text{C}-\text{CH}_4$ that represent CH_4 produced by CO_2 reduction [Whiticar *et al.*, 1986; Whiticar, 1999] may not accurately represent CH_4 from northern terrestrial wetlands. As pointed out by Waldron *et al.* [1999] the CO_2 reduction field in these plots was primarily determined from marine samples where the $\delta\text{D}-\text{H}_2\text{O}$ is near 0‰. Methane produced by CO_2 reduction in terrestrial northern wetlands where the $\delta\text{D}-\text{H}_2\text{O}$ is $\ll 0$ ‰ will clearly not fall in the CO_2 reduction fields as defined by Whiticar *et al.* [1986, their Figure 12] and Whiticar [1999, his Figure 4]. The $\delta\text{D}-\text{CH}_4$ from the CO_2 reduction field in the Whiticar figures ranged from -150 ‰ to -250 ‰ for

$\delta\text{D}-\text{CH}_4$ and -60 ‰ to -110 ‰ $\delta^{13}\text{C}-\text{CH}_4$ (our Figure 7). However, the δD and $\delta^{13}\text{C}$ values for CH_4 from Turnagain Bog ranged from -308 ‰ to -350 ‰ and from -64 to -73 ‰ respectively (Figure 7). At Kings Lake Bog in Washington, USA, $\delta\text{D}-\text{CH}_4$ isotopic values ranged from -276 ‰ to -331 ‰ [Lansdown *et al.*, 1992]. In both of these systems, incubation studies using field samples have shown that CH_4 production is dominated by CO_2 reduction and these incubation studies have been interpreted to represent field conditions [Lansdown *et al.*, 1992; Hines *et al.*, 2001; Duddleston *et al.*, 2002]. Methane isotopic values from several other systems where CO_2 reduction has been inferred to be the dominant mechanism of production yield similar results. Methane from raised bogs in the Glacial Lake Agassiz peatland in northern Minnesota had CH_4 isotopic values of from -260 ‰ to -290 ‰ and -65 to -70 ‰ [Chasar *et al.*, 2000a]. Bleak Lake Bog in Alberta CH_4 values were from -300 ‰ -330 ‰ and -70 to -78 ‰ and are also clearly outside of the range of $\delta\text{D}-\text{CH}_4$ values ascribed to CH_4 from CO_2 reduction (Figure 7a), although the $\delta^{13}\text{C}$ data fit the model.

[25] We assert that $\delta\text{D}-\text{CH}_4$ values cannot be associated with CH_4 production mechanism in an absolute sense. The production mechanism effect is combined with the effect of the δD of H_2O . Note that Glacial Lake Agassiz Fen surficial samples [+ symbols, Chasar *et al.*, 2000a, depths above 1 m] do fall within the bacterial methyl fermentation field and are located to the lower right of (^{13}C enriched and δD depleted) Agassiz bog samples as discussed by Chanton *et al.* [2005]. Acetate fermentation is thought to be greater in the surface soils of fens where the organic matter is more labile [Chanton *et al.*, 2005].

[26] However, if we shift the δD values of these northern wetland CH_4 samples by the value of the environmental water where they were collected (Figure 7b),

$$\begin{aligned} \delta\text{D}-\text{CH}_4 \text{ measured} - \delta\text{D}-\text{H}_2\text{O measured} \\ = \delta\text{D}-\text{CH}_4 \text{ on Figure 7b} \end{aligned} \quad (3)$$

then the values fall within the CO_2 reduction field. Essentially we have corrected them to values they might have had if they had been produced from $\delta\text{D}-\text{H}_2\text{O} = 0$ ‰.

[27] In conclusion, the results of this study and results of field studies compiled by Chanton *et al.* [2005] have provided strong evidence that CH_4 produced from acetate fermentation is relatively ^{13}C enriched and δD depleted relative to CH_4 produced from reduction of CO_2 , if precursor isotopic values are relatively constant. Laboratory incubation studies have produced similar results for $\delta^{13}\text{C}$; CH_4 produced from acetate is ^{13}C enriched relative to CH_4 produced from CO_2 reduction [Sugimoto and Wada, 1993; Waldron *et al.*, 1998a]. The laboratory studies have not produced the apparent δD variation with production mechanism observed in field studies. However, the time span represented by the vertical profiles [Hornibrook *et al.*, 1997; Chasar *et al.*, 2000a, 2000b] where antipathetic variation of $\delta^{13}\text{C}-\text{CH}_4$ and $\delta\text{D}-\text{CH}_4$ have been observed, represent much longer timescales (hundreds of years) than can be represented in laboratory incubations which simulate changes in organic matter quality. One must consider that it may take 500 years to generate a pathway shift through increased recalcitrance of organic matter in a

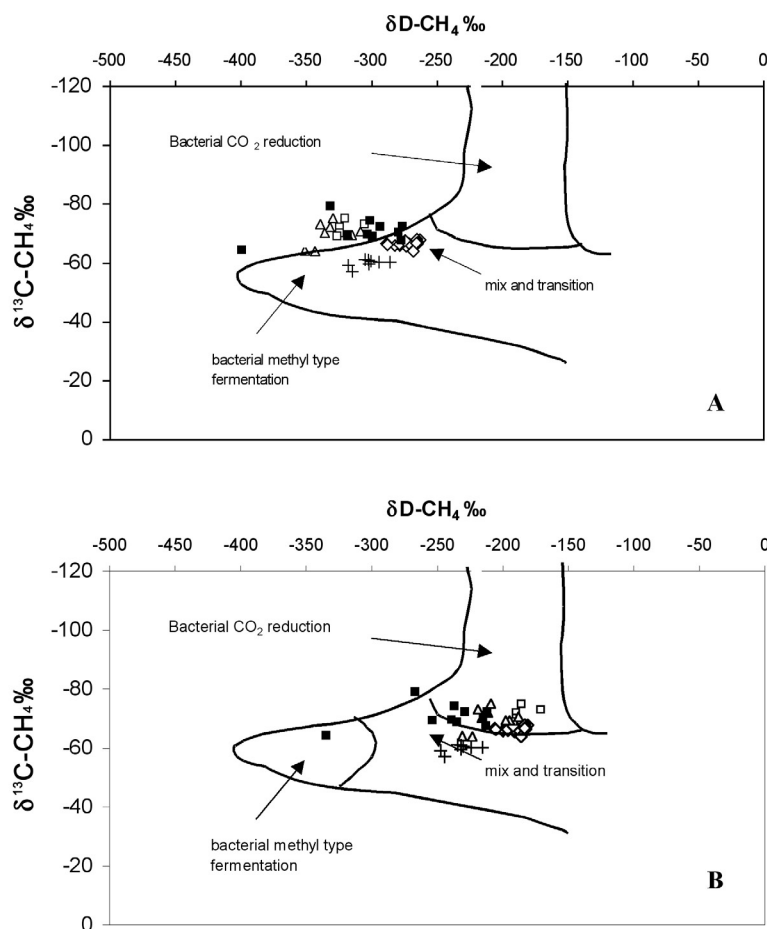


Figure 7. (a) Diagram for classification of CH₄ by a combination of $\delta^{13}C-CH_4$ ‰ and $\delta D-CH_4$ ‰. Data symbols were placed from northern North American *Sphagnum*-dominated bogs where CO₂ reduction dominates CH₄ production. Glacial Lake Agassiz raised bog in northern Minnesota, United States [Chasar *et al.*, 2000a] (open diamonds); Bleak Lake Bog in northern Alberta, Canada [Chanton *et al.*, 2005] (open squares); Turnagain Bog, Alaska, United States (open triangles); Kings Lake Bog in Washington State, United States [Landsdown *et al.*, 1992] (filled squares). While the samples fall toward the bacterial CO₂ reduction field, they are generally δD depleted. This depletion is due to the difference in $\delta D-H_2O$ between terrestrial water at northern latitudes and marine waters. Methane derived from CO₂ reduction in marine samples was used to define the CO₂ reduction field in this figure. Note that Glacial Lake Agassiz Fen samples (+ symbols, Chasar *et al.* [2000a]) do fall within the bacterial methyl fermentation field and are to the lower right (¹³C enriched and δD depleted) relative to Agassiz bog samples [see Chanton *et al.*, 2005]. (b) Similar to Figure 7a except that all $\delta D-CH_4$ data have been adjusted for the $\delta D-H_2O$ of the water at the location where the CH₄ was formed (equation (3)). Glacial Lake Agassiz raised bog (open diamonds) +82‰ [Chanton *et al.*, 2005]; Bleak Lake Bog in northern Alberta, (open squares) +134‰ unpublished data; Turnagain Bog, Alaska (open triangles) +120‰; Kings Lake Bog in Washington State [Landsdown *et al.*, 1992] (filled squares) +64‰ [Waldron *et al.*, 1999]; Glacial Lake Agassiz Fen samples (+ symbols, plus 82‰). This adjustment effectively normalizes the formation water to be 0‰, similar to marine waters. Adapted from Whiticar [1999] and Whiticar *et al.* [1986] with permission from Elsevier.

natural system (E. Hornibrook, personal communication, 2006). So in that respect, incubation studies do not totally rule out the notion that CH₄ production mechanism could be caused by long slow changes in organic matter reactivity.

[28] Burke [1993] noted that CH₄ produced in laboratory incubations and in the rumen with elevated H₂ concentrations exhibit depleted δD values relative to values found in wetlands and sediments. He hypothesized that fractionation

between the $\delta D-H_2O$ and the $\delta D-CH_4$ was related to the partial pressure of H₂. It is possible that variation of $\delta D-CH_4$ associated with variations in H₂ concentration may be an additional factor complicating comparisons of field samples with laboratory incubations. Valentine *et al.* [2004] clearly demonstrated that variations in microbial growth rate can have a huge effect on $\delta D-CH_4$ and postulated that this factor might explain the results of Sugimoto and Wada [1995].

Carrying this idea further, it is interesting to speculate that the variation in δD that has been observed in field studies [e.g., Chanton *et al.*, 2005; Hornibrook *et al.*, 1997; this study] and attributed to production mechanism variations may have been due to covariation of H_2 concentration or microbial growth rate with production mechanism, or some other factor that covaries with production mechanism [c.f. Waldron *et al.*, 1998b]. Perhaps H_2 concentration is greater in soils and sediments with more labile organic matter where acetate fermentation is of more relative importance [Burke, 1993]. Overall, there is a clear need for additional investigation of H isotope fractionation between organic matter, acetate, CH_4 , H_2 and H_2O .

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